

Alex Huth / Seconds 0

TOOLING

GRANT ALLOCATED \$10,000

One-Sentence Description

Automated pipeline for translating Soviet research papers into English.

Proposed Project (+ incubated within RRF)

Alex is building a software or technical pipeline for translating Soviet research papers, adapted from the infrastructure he built previously with [Chinarxiv.org](https://chinaxiv.org). Since joining the Analogue Discord in late February, Alex has created a validated translation pipeline with source-aware harvesting, Cyrillic-aware quality gates, language-aware retry logic and visibility controls.¹ The current pipeline involves automated OCR, translation of Russian technical language, translation of mathematical equations and diagrams, and pdf generation. Alex is also building a website for the database of translated papers, with chat features to “talk” to papers and do advanced search.

Alex has been using available journal papers from math-net.ru, with the first translation batch on *Doklady Akademii Nauk* (Proceedings of the Russian Academy of Sciences). The pipeline currently has a cost estimate at ~\$0.025 per page at batch rates.

The \$10,000 grant was given solely to support Alex’s R&D. A further Soviet research revival project, using Alex’s pipeline, is being incubated within Revival. I am working with historians to identify and curate high-value journals to begin first batches of translations for public publishing.

The future of this project can go multiple ways:

- 1) **Guest Archivist Collaboration.** Having archivists, PhDs and history professors be guest curators of archives, and write a short introduction on their selection of high impact journals or papers
- 2) **Translation Package as Service/Product.** People can purchase archive bundles from us to translate. E.g. research bundle on peptides.

Alex has expressed consistent interest in working with us to figure out the suitable forms of research sharing in the future.

Revival Relevance

Alex came to Revival out of a shared interest and mission in democratizing research and open publishing.

We at Revival identify a huge need in revisiting Soviet research archives, which needs to begin with an updated, accurate translation pipeline. Hence it was immediately obvious to us that we should support Alex’s work on tooling, and work with him more expansively on the bigger project of recovering Soviet research.

As English became the globalized language in scientific knowledge production in the 20th century, Russian journal papers became increasingly marginalized and the research results forgotten. And while the NSF

¹ Validation on translation quality was conducted by Ulkar, who volunteered to help with the broader Soviet research revival project.

funded translations of major Soviet journals during the height of the Cold War, the brute-force translation efforts during that time prioritized raw data, speed and military intelligence, with the resulting English prose prone to being plagued with errors. In addition, in a rush to train American scientists to read Russian, educators promoted a simplified dialect, “Scientific Russian”, which acted as a reduced communications code and essentially functioned as a pidgin.² All factors combined reduce legibility of the original research in its transmission from the USSR to today’s scientific context across the temporal and cultural registers.

With Alex’s translation pipeline, we are excited to redo the translation of journals (identified by “guest archivists” we invite, involving academic historians as well as passionate entrepreneurs) with more precise language, and in a more curated manner, surfacing “alpha” papers that could provide tangible values to research today.

Wendi’s Notes

1. Alex previously built and runs chinaxiv.org

[Chinaxiv.org](https://chinaxiv.org) is a live open-access platform serving 22,700+ machine-translated Chinese research papers with full text, search, and a free public API. Alex’s Revival project takes the same core infrastructure and adapts to the Russian academic journal context.

Alex published [an article about his mission](#) on *Reason* in early May, in which he outlined problems of the peer review system, and how it structurally encourages fraudulent and extractive behaviors damaging research itself. Alex argues that AI can play an instrumental and supporting role in science research, and explains how [Chinaxiv.org](https://chinaxiv.org) came from realizing one day that thousands of Chinese preprints are untranslated with no real reference.

2. Alex has a wealth of related experience

Alex previously digitized files and records for the Air Force. He has had extensive hands-on experience with handling physical records for several years. This is exciting for us as we continue building on the Soviet translation project together.

3. Alex is aligned with our mission and we are interested in more long-term collaborations.

Alex has been an active participant in the Analogue Discord community since joining us in late February. He has kept us updated on the pipeline R&D, while working with Ulkar, a native Russian speaker and scientist we connected Alex with, to improve the translation quality. Alex’s *Reason* article and previous experience show an important commitment to open science publishing. We are interested in exploring long-term collaboration with Alex and supporting his Chinaxiv project, and other projects along similar ethos of accessible publishing, archival technology, and knowledge translation and sharing pipelines.

² Michael Gordin, *Scientific Babel*.